

Supervisor Dr. Hesham Alhumyni
Hetaf Al-qumsani, Hatoun Al-qumsani, Esraa Althomali, Shaimaa Alzahrani, Rana Al-harhi, Atheer Al-malki

Reasons

This project helps to reduce many problems within slums such as visual distortion. It also helps to solve the problem of violating buildings



Abstract



The Kingdom suffers from slums and visual distortion, and municipalities are trying hard to solve this problem. Because of the expansion and proliferation of these areas, it is difficult for human resources to resolve them in record time. Citizens' lack of awareness of the destruction of the urban landscape of cities and neighborhoods caused by such abuses. To address this problem, we will design a drone to identify visual distortions such as general hygiene, drilling barriers, street excavations, lampposts, rundown pavements, and construction residues by locating, filming. so that its image and location are uploaded. Organize neighborhoods by identifying abuses in the streets without using human resources. This drone is characterized by the organization of random neighborhood's, accuracy in identifying the of fence and its location and scanning a greater distance in a shorter time to accomplish tasks.

The Future Vision Of The Project



It is possible to design a robot in the future linked to the drone so that the drone will photograph excavations in the streets, lampposts, etc. and send it to the robot, and the robot will fix it. .

The Appropriate Institution To Adopt Our Project



Ministry of Municipal & Rural Affairs & Housing

Added Value In The Project

Determine the violating buildings with using YOLO: Real-Time Object Detection system



Objectives

This project aims to achieve the following objectives:

- Organize randoms using YOLO Detection to identify domestic-based violations and abuses.
- detect all visual distortions to remove them.

Conclusion:

In this research, we designed and developed a system for features to lessen the visual distortion . Additionally, we may organize neighborhoods and lessen street encroachment. Solving these issues and lowering uncivilized violations are crucial

